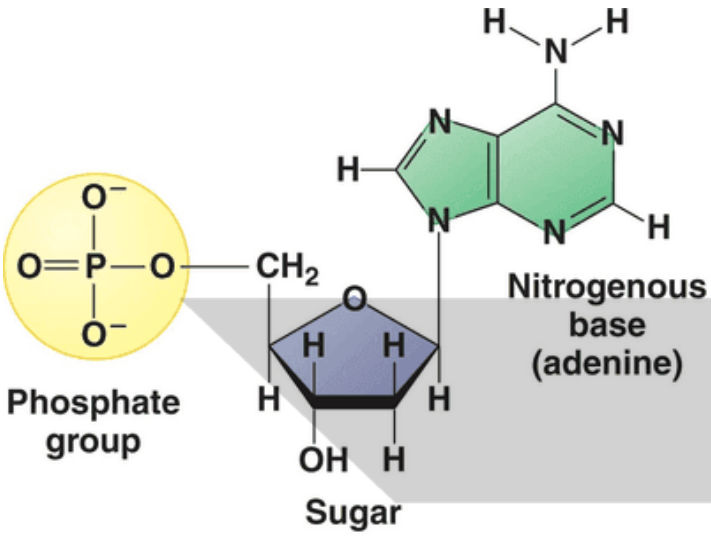
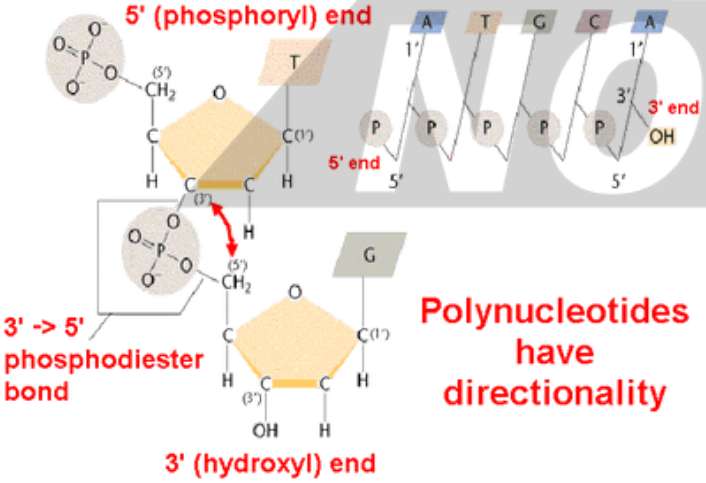
1. Nucleic Acid & Protein Synthesis

1.1. Structure of Nucleotides and ATP

Nucleotide



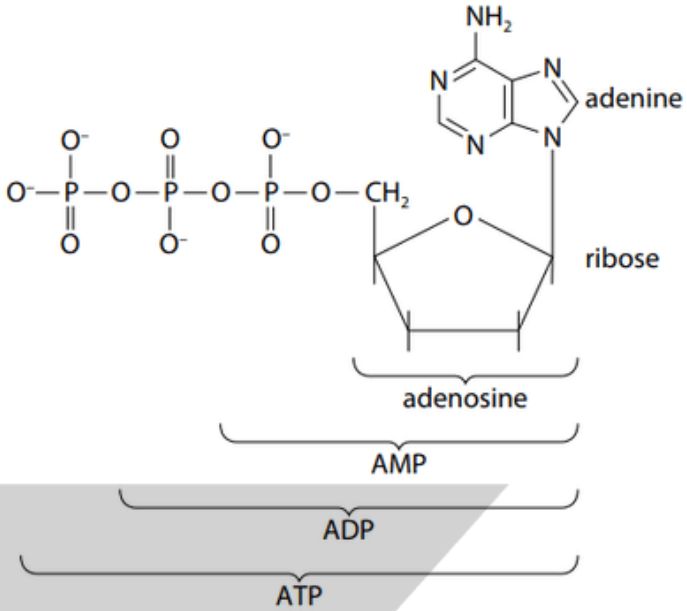
- Basic building blocks of nucleic acids, such as DNA and RNA
- Organic compound of a nitrogenous base, a pentose sugar, and a phosphate group
- Two nucleotides can be joined together by a condensation reaction to form dinucleotides
- Many nucleotides are linked into a long chain, forming a polynucleotide, either DNA or RNA (This happens in the nucleus during interphase)
- The covalent **sugar–phosphate ester bonds (phosphodiester bonds)** link the 5-carbon of one sugar molecule and the 3-carbon of the next
- The polynucleotide strand is said to have **3′and 5′ends**



(after Klug & Cummings 1997)

ATP

- **Adenosine Triphosphate** (Adenosine = Adenine + Ribose sugar)
- Three components: Adenine (nitrogenous base), Ribose (sugar), phosphate
- Phosphorylated nucleotide



Nitrogenous Bases

- **Purines** are nitrogenous bases with double-ring structures (adenine and guanine)
- **Pyrimidines** are nitrogenous bases with single-ring structures (thymine, cytosine, uracil)
- **Complementary base pairing** between adenine and thymine (or adenine and uracil in RNA) and between guanine and cytosine
- Adenine and thymine (or uracil in RNA) form 2 hydrogen bonds, and guanine and cytosine form 3

1.2. DNA Structure and Replication

- DNA comprises two antiparallel polynucleotide strands lying side by side, held together by hydrogen bonds
- One strand is in the 3′ to 5′ direction, and the other is in the 5′ to 3′ direction, hence **antiparallel.**
- It has 4 nitrogenous bases: adenine, thymine, guanine, and cytosine
- The phosphodiester bonds make up a sugar-phosphate backbone, which supports the shape of the DNA
- Hydrogen bonds between the complementary bases hold the DNA **double helix** together, maintain 3D structure, and ensure stability

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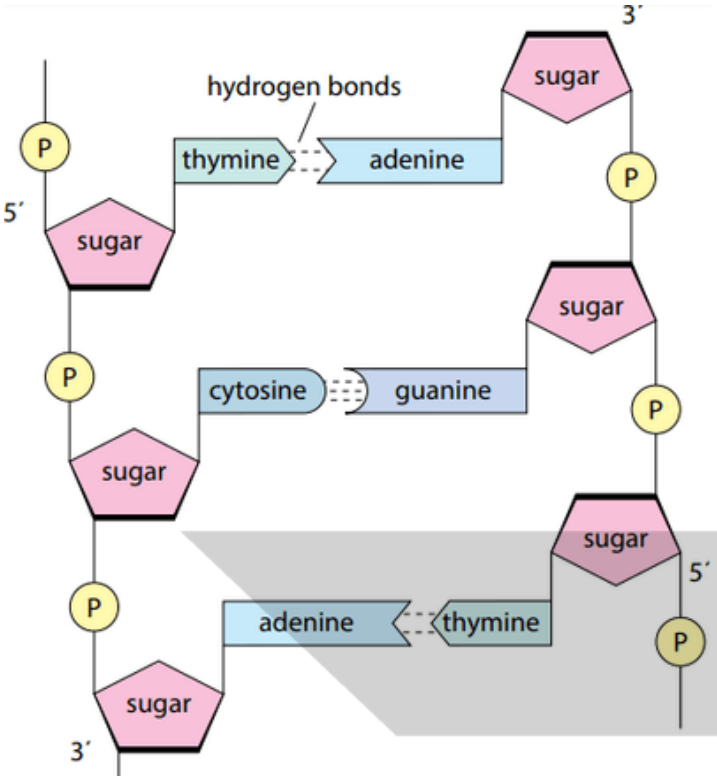
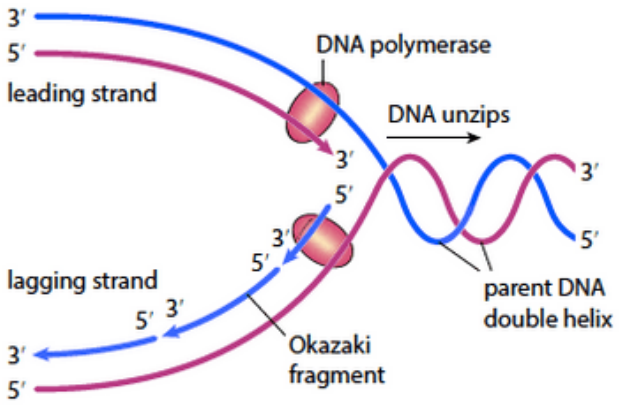


Replication of DNA:

- Occurs during interphase (S phase)
- Each new DNA molecule contains one old strand and a complementary new one (hence, replication is called **semi-conservative**)
  - The DNA double helix unwinds and 'unzips' as the hydrogen bonds between the base pairs break by the DNA helicase enzyme.
  - Both strands are used as templates.
  - DNA polymerase catalyses the synthesis of the phosphodiester backbone by linking adjacent nucleotides
  - DNA polymerase can only add nucleotides in the 5′ to 3′ direction
  - It uses the 3′ to 5′ strand as a template and forms the polynucleotide in a 5′ to 3′ direction
  - This strand is continuously synthesized (Leading strand)
  - The new strand synthesized in 5′ to 3′ is made in small Okazaki fragments (Lagging strand)
  - The DNA ligase enzyme joins Okazaki fragments together by forming phosphodiester bonds

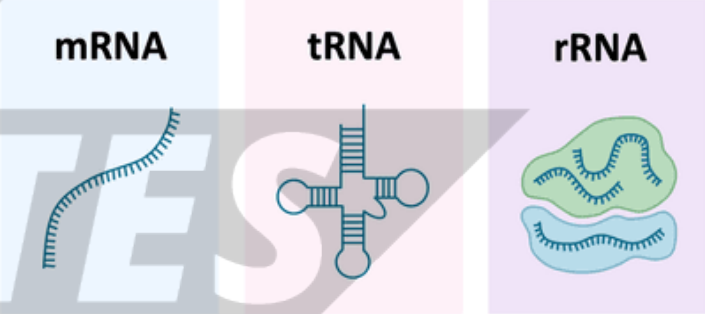


1.3. RNA

- RNA is a **single-stranded polynucleotide chain**
- It contains a pentose sugar (ribose)
- Has 4 nitrogenous bases: **adenine, uracil, guanine and cytosine**

There are different types of RNA, which include:

- **mRNA (messenger RNA):** Carries the genetic information as a template from the nucleus to the ribosome for translation, remains as an unfolded strand
- **tRNA (transfer RNA)**: Has a specific amino acid at one end and an anticodon at the other; it fits onto the ribosome mRNA at a complementary codon for protein synthesis, folded into complex structures
- **rRNA (ribosomal RNA):** Primary component of ribosomes, synthesised in the nucleolus, folded into complex structures



1.4. The Genetic Code

Gene:

- A sequence of nucleotides that forms part of a DNA molecule
- A polypeptide is coded for by a gene

Codon:

- Each group of three bases in mRNA constitutes a codon
- There are 4 nucleotide bases, 4 x 4 x 4 = 64 possible amino acids. However, only 20 amino acids exist; hence the genetic code is:
  - **Redundant** or **degenerate** - Multiple codon codes for the same amino acid

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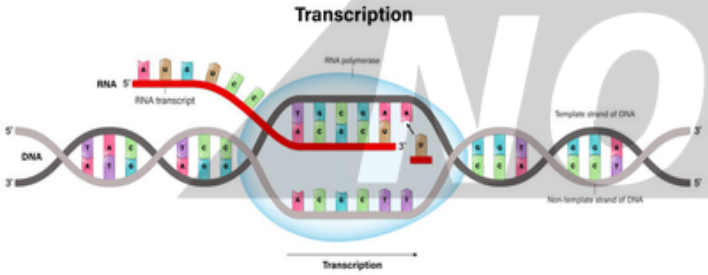
- The genetic code is **universal** - All organisms use the same code
- It has start and stop codons to mark the beginning and end of the gene for protein synthesis

first letter	second letter: U	second letter: C	second letter: A	second letter: G	third letter
U	UUU, UUC } Phe; UUA, UUG } Leu	UCU, UCC, UCA, UCG } Ser	UAU, UAC } Phe; **UAA Stop**; **UAG Stop**	UGU, UGC } Cys; UGA **Stop**; UGG Trp	U, C, A, G
C	CUU, CUC, CUA, CUG } Leu	CCU, CCC, CCA, CCG } Pro	CAU, CAC } His; CAA, CAG } Gln	CGU, CGC, CGA, CGG } Arg	U, C, A, G
A	AUU, AUC, AUA } Ile; **AUG Met**	ACU, ACC, ACA, ACG } Thr	AAU, AAC } Asn; AAA, AAG } Lys	AGU, AGC } Ser; AGA, AGG } Arg	U, C, A, G
G	GUU, GUC, GUA, GUG } Val	GCU, GCC, GCA, GCG } Ala	GAU, GAC } Asp; GAA, GAG } Glu	GGU, GGC, GGA, GGG } Gly	U, C, A, G

1.5. Protein Synthesis

Transcription

- Occurs in the nucleus
- DNA unwinds to form two strands (template and non-template)
- Only one strand acts as the template (the template strand)
- Free-activated RNA nucleotides line up with their complementary base and form H-bonds
- **RNA polymerase** catalyses the synthesis of phosphodiester bonds to form a sugar-phosphate backbone
- Hydrogen bonds between the DNA and mRNA strands are then broken
- DNA is reformed
- mRNA strand then leaves the nucleus through the nuclear pores

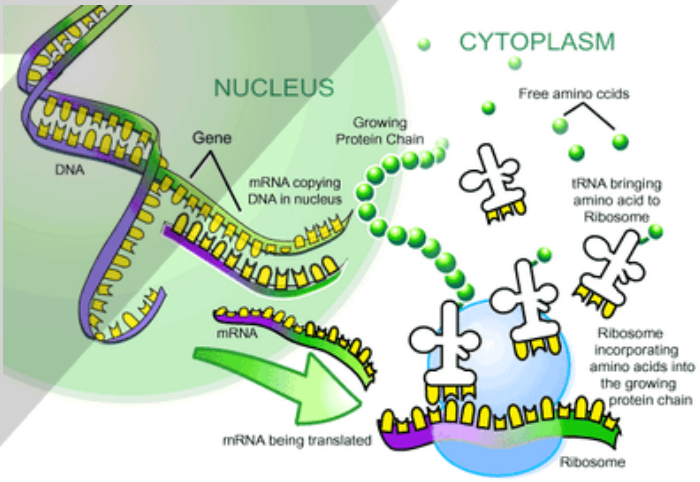


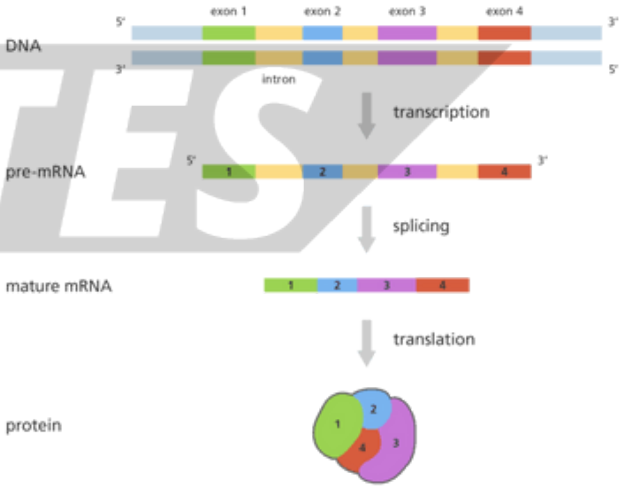
Post-transcriptional RNA modification

- Eukaryotic genes are made of introns and exons
- **Exons:** Coding sequence
- **Introns:** Non-coding sequence which is not translated
- Introns are removed from the primary transcript, leaving behind only exons, which will undertake translation (this process is called RNA splicing)
- Primary transcript: Original molecule of RNA before splicing

Translation

- Small ribosomal subunit attaches to mRNA
- tRNA enters the ribosome and attaches to the mRNA
- A codon on the mRNA attaches to a specific anticodon on the tRNA
  - AUG is the start codon; complementary anticodon is UAC, which brings the amino acid methionine
- Only 2 tRNA molecules can fit in the ribosome at the same time
- Each tRNA carries a specific amino acid
- A peptide bond is formed between the amino acids of 2 adjacent tRNA molecules (at any time, 2 tRNA molecules can fit in the ribosome)
- The ribosome moves along the mRNA, reading the next codon
- Another tRNA molecule brings a third amino acid, which joins the previous one
- The first tRNA leaves and is reused
- The polypeptide chain grows until a 'stop' codon: UAA, UAG or UGA is reached.





1.6. Mutation

Gene Mutation

- A change in the sequence of base pairs in a DNA molecule that may result in an altered polypeptide
- Caused by mutagens like radiation, tar, tobacco, etc.

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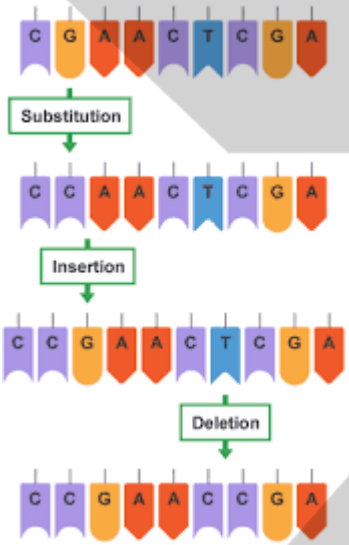
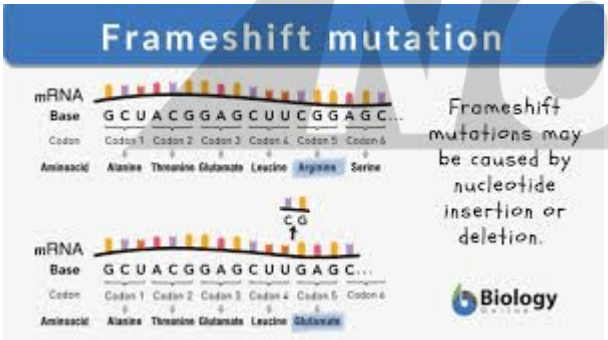
- Random events
- Harmful as it changes the amino acid sequence, which changes the mRNA created
- This further alters the tRNA that attaches to the ribosome, changing the amino acid brought, and hence alters polypeptide formation

Types of mutations

- **Substitution mutation:** A nucleotide base is replaced by a different nucleotide base
  - E.g., Sickle cell anaemia, where the base thymine is replaced with adenine (CTT → CAT) so Glutamine becomes Valine
- **Elimination (deletion) mutation**: Removal of one or more nucleotides and is not replaced
- **Insertion mutation:** A nucleotide base is added



- **Frame-shift mutation**
  - Insertion or deletion of one or more nucleotides which results in incorrect reading of the sequence of triplets in the genetic code due to a shift in the reading frame



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